The Box in Balance

Feeling gravity with an accelerometer: tilting, balancing and reading three numbers

Level: Junior — ages 7–10 (Romanian school years II–IV) • **Time in class:** 15–20 min, one 50-minute lesson, or 3 lessons in a week • **Sensors:** MPU-6500 (accelerometer)



The activity screen. The readings in the picture are simulated; your own values will differ.

Before you start

Who it is for — Junior — ages 7–10 (Romanian school years II–IV). It also works in year V as revision, using the calculation task from the project route; that is why the activity also appears in the **Explorer** level's list.

Assumed knowledge — Reading a decimal number and comparing two numbers. Negative numbers are not required: a minus sign simply means "the other way round".

Time in class — Short: 15–20 min. Full: one 50-minute lesson. Project: three 50-minute lessons in the same week (2 h 30 min).

Teacher preparation — 5 min — switch the box on, check the connection, print the sheets.

Sensor settling time — Under 5 seconds. The accelerometer responds immediately; it has no warm-up time, unlike the CO_2 sensor. The "count to three" wait is for the hand to settle, not for the sensor.

Sensors and boxes — MPU-6500. One box is enough for the whole class if you work in a circle; ideally, one box per 4–6 children.

Three routes

Short	The app's seven steps: written prediction, flat reading, a second prediction, tilting and the balance challenge, measuring twice, one conclusion.
Full	The short route, unhurried, plus all six positions and the discussion.
Project	The full lesson, then the ramp, then the corridor and the class display.

1. Where this fits in the curriculum

This is a first encounter with an invisible force made visible as a number on a screen. Children already know that things fall; here they discover that a small chip can tell how much of that falling-force points along each of three directions, and that the numbers change predictably when the box is turned. The activity fits:

- **Natural sciences** (*Științe ale naturii*, Romanian school years III–IV) — forces and motion, gravity as a pull towards the Earth, pushes and pulls, the idea that "up" and "down" are set by the planet, not by the room. In years I–II the same ideas can be touched on, more simply, within **mathematics and exploring the environment** (*Matematică și explorarea mediului*).
- **Mathematics** (*Matematică*, a subject in its own right from year III) — reading and comparing decimals, negative numbers met in a physical setting (a minus sign simply means "the other way round"), estimating, ordering values.
- **Geography** (*Geografie*, year IV only, where the subject appears in the Romanian national timetable) — direction, orientation, the meaning of the words horizontal and vertical.
- **Cross-curricular** — careful observation, recording what you see rather than what you expected to see, taking turns with shared equipment, and a first idea of what a sensor is: why a phone screen rotates when you turn it.

The core skill is **linking a physical action to a change in a number**. Children tilt the box themselves and watch a value respond immediately. That tight loop between gesture and reading is what builds the intuition; the vocabulary (axis, gravity, sensor) can follow afterwards.

2. Time plan

The in-app activity is short — seven steps and about twenty minutes of screen time. It works as a lesson starter, as a full lesson, or as the first of three lessons that grow into a small investigation.

Route	Time	What happens
Short	**15–20 min**	The introduction (2), the first prediction (2), the flat reading (4), the second prediction (1), tilting and the balance challenge (7), **How sure are we?** (2), the conclusion (2). These are exactly the seven steps in the app and they fit into the opening of a lesson.
Full	**50 min**	The short route, unhurried, with every child getting a turn to tilt the box, with all six positions entered in the table on the sheet, and with the discussion of what the sensor actually measures — why the table pushes up and what the box would read in free fall.
Project, lesson 2	50 min	The ramp investigation: the box sits on a board raised in steps (one book, two books, three books) and the class records how AX or AY grows while AZ shrinks. This is where the controlled variable appears.
Project, lesson 3	50 min	The corridor walk and the "carry the box without waking it" challenge: children walk, stop, turn and run while a partner watches the values, then everyone contributes to a class display.

The project route begins with the full lesson and adds lessons 2 and 3, so it occupies **three 50-minute lessons in the same week** — two and a half hours of contact time. No homework and no preparation at home is asked of the children.

One box and thirty children

The most common practical problem in this activity is not the physics, it is the queue. With a single box, sit the class in a circle, project the laptop screen, and pass the box from hand to hand so that each child makes one named move ("tilt it to the right") while everybody else watches the numbers and says out loud what changed. Thirty children, each with a three-second move, takes under two minutes and nobody stands idle. Keep the free exploration for group rotations, if you have a second and a third box.

3. Materials

What you need	Per group	Notes
MagicBox with MPU-6500	1	The accelerometer is on the main board; no extra sensor needs plugging in.
Laptop with the app	1	Connected by USB cable, or over WiFi if your box is configured for it. A short cable restricts the tilting, so allow slack.
Flat, stable table	1	The first reading must be taken on a genuinely level surface. A wobbly table produces a jittery AZ and confuses the class.
Sticky labels or masking tape	3	Mark the edges of the box X, Y and Z before the lesson. Children cannot reason about an axis they cannot see.
Flat board or stiff tray	1	Roughly A4 or larger, to act as the ramp in lesson 2 of the project route. A hardback book cover works.
Hardback books of equal thickness	3–4	Stacked under one end of the board to raise it in equal steps.
Ruler	1	To measure how high the raised end of the board sits.
Protractor (optional)	1	For the stretch group, to compare a measured angle with the one the numbers imply.
Non-slip mat or cloth	1	Under the ramp, so the board does not shoot off the table.
Clipboard and pencil	1 per child	The sheet is filled in standing up.
Small cushion or folded jumper	1	On the floor beside the table; a dropped box lands on it rather than on tiles.

Preparation the day before: switch the box on, open the activity and check that AX, AY and AZ show sensible values with the box lying flat — roughly zero, zero and **plus one**. If AZ reads 0.94 rather than 1.00, that is the sensor's own offset and it is perfectly normal, but it is worth knowing the number in advance so that you are not taken by surprise in front of the class. Stick the X, Y and Z labels on the box, decide which edge will be "the front", and charge the laptop. Clear a table so that the box can be tilted through ninety degrees without knocking anything over.

4. Scientific background for the teacher

What an accelerometer actually measures

A MEMS accelerometer such as the MPU-6500 contains a tiny silicon body — a proof mass — held by springs. When the chip accelerates, the body lags behind and the resulting deflection is read as a change in capacitance. Crucially, the device measures **proper acceleration** — acceleration relative to free fall — and not acceleration relative to the ground.
This is why a box sitting perfectly still on a table does not read zero. The table top pushes up on the box with a force that exactly balances gravity, and it is that upward push the sensor feels. Written for the vertical axis,

$$a_z^{\text{measured}} = \frac{N}{m} = g \approx 9.81 \text{ m/s}^2 \equiv 1\, g.$$

A box in genuine free fall reads 0 on all three axes — which is exactly why a dropped phone knows it is falling. The reading does not say "how fast I am moving"; it says "which way is down, and how hard is something holding me up".

4.1 Why the three numbers always add up the same way

When the box is still, the three readings are the components of a single vector of fixed length, pointing away from the centre of the Earth. Whatever the orientation,

$$\sqrt{a_x^2 + a_y^2 + a_z^2} = 1\, g.$$

This is the quiet, powerful fact underneath the whole activity: gravity is not shared out arbitrarily between the axes, it is *projected* onto them. Tilt the box and one axis gains exactly what another loses, in the Pythagorean sense. Older or faster children can be shown this directly: let them square their three recorded numbers and add them; the answer comes out close to 1 even for a messy, arbitrary tilt. That moment always goes down well.

4.2 How the readings become an angle

The tilt of the box away from horizontal follows from the vertical component:

$$\theta = \arccos\left(\frac{a_z}{g}\right),$$

and the two conventional aircraft-style angles are

$$\text{pitch} = \arctan\left(\frac{a_x}{\sqrt{a_y^2 + a_z^2}}\right), \qquad \text{roll} = \arctan\left(\frac{a_y}{\sqrt{a_x^2 + a_z^2}}\right).$$

You do not teach these formulae to seven-year-olds — the cosine and the other trigonometric functions arrive only in year VII —, but it is worth knowing them, because they explain something the children will notice: near the flat position the readings change only slowly with tilt (the cosine varies very slowly around zero), while near ninety degrees a small movement produces a large change. The sensor is *least* sensitive to tilt exactly where the children start — which is why the first tilts need to be generous.

4.3 The 0.577 puzzle in the last step of the app

The last step of the app asks what it means when $a_x = a_y = a_z = 0.577\, g$. The answer is that $0.577 \approx 1/\sqrt{3}$, and

$$3 \times \left(\tfrac{1}{\sqrt{3}}\right)^2 = 1,$$

so the total is still exactly one g. Physically, the box is balanced on a corner with its body diagonal pointing straight down: gravity is shared into three equal parts. Children can be invited to hunt for this position themselves, by trying to make all three numbers equal — a far better exercise than being told the answer.

4.4 Static versus dynamic readings

Everything above assumes the box is still. As soon as it is shaken, tapped or swung, the sensor adds the acceleration of that motion on top of the gravity component, and the numbers jump about wildly — values well beyond $\pm 2\,g$ are easy to produce. From firmware 2.2.5 the chip smooths its own readings at about 20 Hz: hand tremor shows less, but a sharp tap still shows. This is not a fault. It is exactly the signal a car uses to decide that a crash has occurred. It is worth saying that the sensor cannot, by itself, tell "tilted" from "accelerating sideways". That ambiguity is why real devices combine the accelerometer with the gyroscope that sits in the same MPU-6500 chip.

5. Lesson flow

The in-app wizard has seven steps: **Introduction: What is an accelerometer?**; a prediction; **Experiment: The box lying flat**; a second prediction; **Experiment: Tilt it in every direction**; **How sure are we? Measure twice**; and **Conclusion: What did we discover?**. **Before each measurement the app asks for a written prediction** in the on-screen journal and quotes it back in the conclusion; the journal is saved to disk by itself. Below is what the teacher does around them. The times are for the 20-minute short route; double them, with ten minutes for discussion, for the full 50-minute lesson.

1. **Introduction on screen (2 min).** The activity is opened from the **Activities** page, with the **Start** button on the card **The Box in Balance**. Work through the first step of the wizard together. Establish the three words the class will use all lesson: *axis*, *gravity*, *sensor*. Point at the labelled edges of the box as you name X (left — right), Y (front — back) and Z (up — down). The button is **Ready to explore!**.
2. **The first written prediction (2 min).** The second step. Ask every child to write or draw on the sheet an answer to the two questions the app also asks: *the box lies still on the table — will the sensor show zero, or something else?* and *if we tilt it towards the window, which numbers will change?* Insist on a committed answer from every child, without exception, and a reason in two words; one of them also types it into the on-screen journal. Do not correct anyone yet. The point of writing is that a prediction on paper cannot be quietly revised later, and that is what makes the surprise stick. The button is **My prediction is written**.

Technical note — right before the first reading

The sensor. MPU-6500, a three-axis accelerometer, on the main board of the box. Nothing extra is plugged in and **there is no calibration step**: the values you see are the ones the chip gives.

What a box lying flat reads. AZ $\approx +1$ g, with AX and AY close to zero. The chip is mounted upside down inside the box, but the firmware already corrects for that mounting (the -1 signs on AX and AZ, that is, a 180° rotation about the Y axis), so the value that reaches the application is the normal, positive one. This is the reading you will see in class and the one to expect from the children.

If you see $-1\,g$ with the box flat on the table, that is not a quirk of the sensor, and it has two possible causes. If the app flagged **Old firmware** when the box connected, the box is running firmware older than v2.2.1: reflash it — the flashing guide is in the `firmware/` folder. If the firmware is current, the box has its sensor fitted the other way up: under **Settings → Box names** tick **Motion sensor upside down** for it, once. With firmware

2.3.0 or newer the tick is kept in the box itself, so it counts on every computer; the box must be connected when you tick it. Do not compensate for the sign in the lesson, and do not explain to the children a minus sign that should not be there.
The cards. On the two experiment steps there are three cards, **Accel. X**, **Accel. Y** and **Accel. Z**, in *g*, with **two decimals at every level** — rounded to whole numbers they would read 0, 0 and 1 and hide the sensor's offset. Beside them sits the **Capture** button: with the box flat and, later, with the box on its edge, it saves the mean of the last few seconds, and the conclusion quotes exactly these values, plus their "all three together" total.
What the chart shows. Below the three cards, the chart has **three bands, one each for AX, AY and AZ**, all visible at once. The chart **keeps gravity**: with the box flat, the AZ band sits at +1, exactly like the value on the card. The time axis shows minutes and seconds from the start of the step.
The buttons you will use. Below the text of each step, the activity moves on with the green button at the bottom right. Its label changes from step to step: **Ready to explore!**, **My prediction is written**, **I read the flat box**, **My prediction is written** again, **I explored every direction**, **Understood** and, at the last step, **Finish**. On the two experiment steps the button stays pale until you press **Capture**, with a tip underneath; you can still move on. The **Back** button, on the left, goes back one step. **At the Junior level the chart has no tool buttons.** At the Explorer level, above the chart sit **Smooth: on / Smooth: off** — leave smoothing on; switched off, the chart shows every tremor of the hand —, **Pause** — which freezes the chart, useful when you want to show it to the class — and **Clear**.
Exporting the data. Two buttons appear on the experiment steps: **PNG** saves the chart as an image, and **Excel** saves the step's raw (unsmoothed) readings to an `.xlsx` file, with an `elapsed_s` column giving the seconds since the step began. **There is no CSV button inside the activity.** The CSV file is taken from the **Data & Export** page of the application, with the **Excel** button on the box's row, choosing CSV in the save window — that is where you go if you want the data in another spreadsheet program. Save the file using the pattern `class_date_activity` — for example `3B_2026-09-14_echilibrul`. Note on the sheet the place (room and table), the time and the conditions (box still or moving), check that the file really was saved before closing the application, and photograph the set-up whenever the position of the box matters.

3. **The flat reading (3 min).** The step **Experiment: The box lying flat**. Place the box flat and still and read the three values out loud. Expect roughly 0, 0 and +1. Ask the class to compare this with their own prediction. The key question is: *the box is not moving at all — so why is one of the numbers not zero?* Accept and value every attempt at an answer, then give the explanation: the sensor feels the table underneath pushing the box up, so that it does not fall. The app now says the same sentence. Press **Capture** and have the children write the three exact values, including the untidy decimals, into the table on the sheet. The green button is **I read the flat box**.

4. **Naming the offset (1 min).** If AZ reads 0.97 rather than 1.00, say so openly and give it a name: every real instrument has a small error, scientists call it an *offset*, and honest scientists write down what the instrument said rather than what it should have said. It is a thirty-second remark that pays back for years.

5. **The second prediction (1 min).** The app asks: *if we stand the box on its edge, where will the one g go? What will AZ show then?* A short written answer, before any tilting.

6. **Guided tilting (4 min).** The step **Experiment: Tilt it in every direction**; conduct it rather than letting it run free. Call one move at a time — tilt right, tilt left, tip forward, stand it on its edge, turn it upside down — and after each one ask a different child to say which number changed and whether it went up or down. Standing it on its edge is the important one:

AZ falls close to zero while another axis climbs towards 1, and gravity visibly moves from one column to another; with the box on its edge, press **Capture**. Upside down, AZ reads about $-1\,g$: the same push, from the other side.

7. **The balance challenge (3 min).** Set the app's challenge: can anyone hold the box so that all three numbers are close to zero? The answer is no, and the children have to discover it by failing. After several failed attempts, ask why. As long as something holds the box — the table or a hand —, that one-g push has to be somewhere; it cannot be hidden from all three axes at once. **The only moment all three would read zero is free fall**, when nothing holds the box at all — and that one we discuss rather than do. If a group is ready for more, send them to the hard but possible version: make all three numbers *equal*. When you have finished, the green button is **I explored every direction**.
8. **How sure are we? (2 min).** The app's uncertainty step, at the Junior level *Measure twice*: the box is laid flat again, in the same place, and the three numbers are read once more, three times. Readings close together mean a good measurement; if they jump about, the box moved — $1.8\,g$ does not mean gravity grew, it means someone shook the box. The button is **Understood**.
9. **Round-up (2 min).** Close on the app's conclusion step, which quotes the prediction and the captured numbers — flat and on its edge — and whose green button reads **Finish**. Make the connection outwards: the phone in your pocket rotates its screen using exactly this sensor, driving games use it as a steering wheel, and a car airbag fires when its accelerometer suddenly sees a very large value. Finish by asking every child to complete the conclusion sentence on the sheet.

The investigation, in one table

The question. Do the numbers change when I turn the box, and how?
The hypothesis. Written on the sheet **before** any measurement, with a reason in a few words, and not changed afterwards.
What I change. The position of the box: flat, tilted right, tilted left, tipped forward, on its edge, upside down.
What I measure. The three readings AX, AY and AZ, in g, to two decimal places.
What I keep the same. The same table, the same box, the same person reading the screen, and, for every reading, the box held still.
How many repeats. Three readings for each position. If the three are close together, the value is good; if they jump about, the box moved.
What makes a reading valid — and what makes it an artefact. The box must be **still**. A moving box adds its own acceleration on top of gravity, and the number on the screen then says nothing about tilt: a reading of $1.8\,g$ does not mean gravity got stronger, it means somebody shook the box. The class rule: move it, stop, count to three, only then read.

6. Analysis after the activity

6.1 What the data should look like

Position of the box	AX (g)	AY (g)	AZ (g)	What it means
Flat on the table	0.00±0.05	0.00 ± 0.05	+0.95 to +1.05	All of gravity sits on the vertical axis.
Tilted right, about 30°	+0.4 to +0.6	near 0	0.85 to 0.90	One axis gains, the vertical loses a little.
Tilted left, about 30°	−0.4 to −0.6	near 0	0.85 to 0.90	Same size, opposite sign: direction matters.
Tipped forward, about 30°	near 0	±0.4 to ±0.6	0.85 to 0.90	Now AY changes, not AX; the sign depends on which edge you called "the front".
On its edge	±0.9 to ±1.0	near 0	0.00 ± 0.1	Gravity has moved completely onto a horizontal axis.
Upside down	near 0	near 0	−0.95 to −1.05	The negative sign is the whole point of this reading.
Balanced on a corner	≈ 0.58	≈ 0.58	≈ 0.58	Gravity shared equally three ways.
Being shaken	anything	anything	anything	Movement is added on top of gravity; values beyond 2 g are common.

The row to remember is the first: a box lying flat on the table gives **AZ ≈ +1 g**, positive. Individual boxes differ from one another by a few hundredths of a g, and the exact angle a child reaches is never repeatable, so do not chase agreement between groups on the second decimal place. What must come out the same everywhere are the *signs*, *which number changed*, and the fact that the values move from one axis to another rather than appearing from nowhere.

6.2 Questions to ask afterwards

1. The box was completely still and yet one of the numbers was nearly 1. What was the sensor feeling, if the box was not moving?
2. When you tilted the box to the right, AX went up and AZ went down. Where did the "missing" part of AZ go?
3. Two children both tilted the box to the right, but one got +0.3 and the other got +0.7. Neither of them made a mistake. What was different?
4. What do you think all three numbers would say if we could let the box fall down a very deep hole, with nothing holding it up?
5. Your phone knows when you turn it on its side. Using what you saw today, explain to someone in another class how it manages that.

6.3 Ideas for assessment along the way

- **Low stakes:** give each child a printed picture of the box in three positions (flat, on its edge, upside down) and ask them to draw an arrow showing which way gravity pulls, then circle the letter that will have the big number.
- **Standard:** the completed sheet — a written prediction, the table of readings for at least four positions, and a conclusion sentence, in the child's own words, about why a still box does not

read zero.

- **Extension:** the ramp investigation, written up as a small table of "number of books" against AX and AZ, with a sentence about the pattern; or squaring and adding the three readings from an arbitrary tilt, to see that the total stays close to 1.

Common pitfalls and misconceptions

"The sensor measures speed." By far the most common idea children bring with them, and it is reinforced by the word *acceleration*. Confront it directly: carry the box slowly and steadily across the room and show that the numbers do not care. Then start and stop sharply, and show that they do. The sensor notices *changes* in motion and the pull of the Earth, never steady speed.

"Zero means nothing is happening." A reading of 0 on AX means that gravity is not pointing along X, not that the box is broken or that the axis is switched off.

"The minus sign means it is wrong." Negative numbers arrive here as a direction, not as an error and not as a quantity smaller than nothing. Say "minus zero point five" out loud, and pair it with the physical gesture, every single time.

Tilting too timidly to see anything. Near the flat position the readings barely move, and children who are being careful with the equipment produce no signal and conclude that nothing is happening. Model a bold, obvious tilt yourself first.

Shaking the box while trying to read it. Excited hands make the numbers dance and nothing can be recorded. Introduce the rule: move it, stop, count to three, read.

Expecting AZ to be exactly 1.000. It will not be, and a child who insists on writing 1.0 "because that is the right answer" has learned the wrong lesson. Praise honest recording.

Chasing the impossible challenge for too long. While the box is being held, all three axes cannot read zero. Let the failure happen, then close the subject within two minutes at most, before frustration sets in.

Safety and classroom management

This is a low-risk indoor activity, but children are handling a cabled electronic device and deliberately turning it over, so a few rules are welcome.

Dropping. The box will be dropped at some point in the lesson. Work over a table, with a cushion or a folded jumper on the floor beside it, keep the box away from the table edge, and say plainly that a dropped box is nobody's fault and must be reported, not hidden. Never let a child hold the box above head height or over another child.

Cables. A USB cable running from the laptop to a box being waved through the air is both a trip hazard and a way of pulling the laptop off the table. Route the cable behind the table, keep the laptop away from the edge, and clear the tilting area of bags and chairs. If your box is configured for WiFi, use it and the problem disappears entirely.

The ramp. In lesson 2 of the project route, put a non-slip mat under the raised board and never let a child stack books higher than about 15 cm. Nothing heavy sits on the raised end, and children keep their fingers out from under the board.

Turn-taking and supervision. Excitement is the main management risk. One named child holds the box at any moment, everyone else keeps their hands down, and between turns the box is placed on the table and picked up from there, not thrown through the air from one child to another. If you are working with several groups, every active box needs an adult or a responsible pupil beside it.

Hygiene and care. The box passes through many hands; wipe it between groups if that is your school's practice. Remind children not to touch the exposed electronics, only the case.

Adaptations for pupils with special educational needs

The single-axis version. Cover two of the three cards on the screen with a sheet of paper and leave only one visible — usually AX. The task becomes a single question: "did the number get bigger or smaller?", recorded with an up or down arrow rather than a figure. Three numbers changing at once are too much for a child who is still building attention; one is enough to show the link between gesture and number.

One box, the whole class. Sit the children in a circle, with the screen projected, and pass the box round: each child makes one named move while the rest watch the numbers. Alternatively, a rotating station — the box stays in one corner of the room and groups come to it for three minutes at a time, while the others work on drawing the positions.

For children with motor difficulties. Not being able to hold the box does not mean not taking part. Three real roles, rotating at every position:

- **screen reader** — says the three numbers out loud so the whole group hears them; this is the role with the greatest scientific value, because it is this child who decides when the box is still enough for the reading to count;
- **position drawer** — sketches on the sheet how the box is sitting for each row of the table, with an arrow for "down";
- **data recorder** — fills in the group's table and checks at the end that no row has been left empty.

The box can be held by a classmate, on the instruction of the child with motor difficulties: "tilt it further to the right" is an experimental decision too, and whoever makes it is running the experiment.

Support with writing and calculation. Decimals are not compulsory. A child may record only the whole number and the sign ("about 1", "about 0", "minus"); for the prediction, they may tick one of the boxes already printed instead of writing a free-form answer. Spoken answers are accepted instead of written ones, and the teacher or a classmate writes them on the sheet. The student sheet comes with its labels and rows already printed for exactly this reason.

Language, sensory needs and anxiety. Use the same three words all lesson (axis, gravity, sensor) and always pair them with the same gesture. For children sensitive to noise, the balance challenge can be done quietly, one at a time, rather than with the whole class shouting. Say in advance that at the balance challenge **everyone will fail**: it is the one moment of the lesson where failure is the right answer, and anxious children need to know that beforehand, not afterwards.

If things do not go to plan

You have one box and thirty children. Circle, screen projected, one named move per child. Thirty three-second moves take under two minutes. The table is filled in together on the board, and the children copy it onto their sheets.

You have twenty minutes, not fifty. Drop the free tilting and the ramp. Keep exactly four moments: the written prediction, the flat reading, standing the box on its edge, and the balance challenge. Fill in only the first three rows of the table. The conclusion is said out loud rather than written.

You have no projector. Sit the class in a semicircle around the laptop, close enough to see the screen, and name a **screen reader** for each position, who says the numbers out loud for everybody. It works just as well, and sometimes better: somebody has to pronounce every value.

The box will not connect. Do not abandon the lesson. Use the "What the data should look like" table from this guide as a printed data set: the children write their predictions first, then receive the real values row by row, and check whether they guessed the sign and which

number changed. The discussion of why a still box does not read zero happens unchanged. Repeat the measurement on another day, after checking the cable, the port selected in the application and, if you are on WiFi, the network.
The numbers will not settle at all. The table is wobbling or the box is being touched. Move it to a solid table, take your hands off it, and read after counting to three.

7. Further resources

Stable, well-established starting points. Search within each site for the topic named — these are root domains rather than deep links, so they will not rot.

- **PhET Interactive Simulations** — `https://phet.colorado.edu` — search for the simulations on forces, gravity and motion; the ramp and forces-in-one-dimension activities let the class explore tilt and the components of gravity on screen, before or after the real box.
- **European Space Agency (ESA)** — `https://www.esa.int` — search for weightlessness, free fall and the parabolic flights; the best possible illustration of what a reading of zero on all three axes really means.
- **NASA** — `https://www.nasa.gov` — search for microgravity and the education resources for primary ages; footage of objects floating past astronauts gives children an unforgettable image of "nothing is pushing me up any more".
- **Institute of Physics** — `https://www.iop.org` — search for primary physics and forces; it contains guidance written specifically about children's misconceptions concerning gravity and falling, which map exactly onto the pitfalls in this guide.
- **STEM Learning** — `https://www.stem.org.uk` — search for forces and sensors for ages 7–11; a large library of downloadable activities that extend this lesson into a wider forces topic.
- **United States Geological Survey** — `https://www.usgs.gov` — search for how a seismometer works; the same measurement principle at a much larger scale, and a natural bridge to the seismology activity in this platform.

8. Assessment

The exit ticket, identical to the one on the student sheet: *one thing that surprised you today, and one thing you would do differently if you repeated the measurement.* It is collected in the last two minutes of the lesson and says more about the class's understanding than any mark.

For a more formal judgement, use the **shared three-strand rubric** — *measurement practice*, *data interpretation* and *scientific reasoning.* At Junior level, the three strands look like this:

- **Measurement practice** — the child waits for the box to be still before reading, and records what the screen shows, including the untidy decimals.
- **Data interpretation** — the child says which number changed and in which direction, and links the sign to a real direction.
- **Scientific reasoning** — the child explains, in their own words, why a still box does not read zero, and compares that conclusion with the prediction written at the start.

The rubric itself, with descriptors for all three levels, is defined once in the *curriculum plan* and is not repeated in the guides.

9. The student sheet

The sheet the children fill in is a separate document: `docs/en/student_sheets/box_in_balance.pdf`. It has two pages — the prediction on the first, the measurement and the conclusion on the second — so it prints double-sided, one sheet of paper per child. It is built on the three layers — what I see on the screen, what I do on my own, the conclusion I write — asks for the prediction **before** any measurement, and comes with the table already labelled and its rows printed. The table is not repeated here precisely so that the two cannot drift apart: there is a single printable version, and it is the sheet.